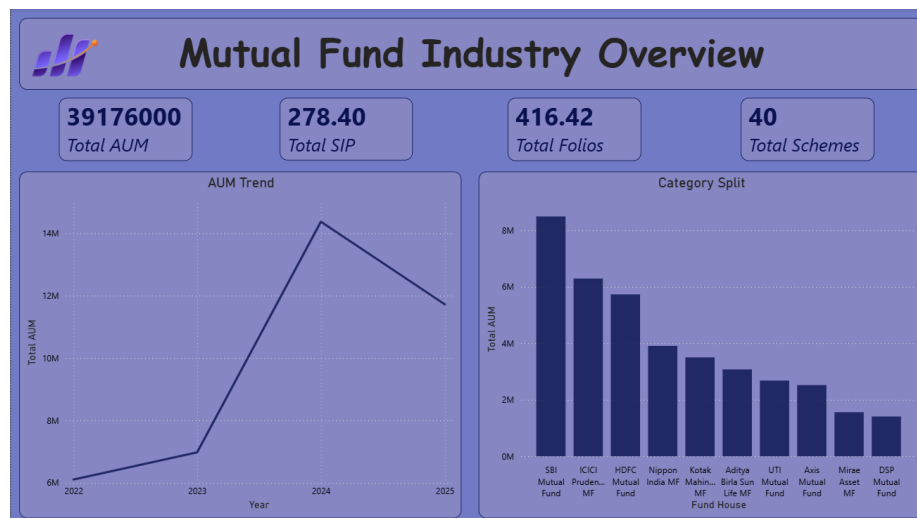


BLUESTOCK

Mutual Fund Analytics Capstone

Final Project Report



End-to-end ETL, SQL analytics, financial performance modelling, risk analytics and an interactive Power BI dashboard across 40 Indian mutual fund schemes

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Executive Summary

This report documents a six-day, end-to-end mutual fund analytics capstone covering data ingestion, cleaning, SQL warehousing, exploratory analysis, performance and risk analytics, an interactive Power BI dashboard, and advanced modelling for 40 actively tracked equity, debt and liquid schemes across 10 Indian AMCs. All figures below are computed directly from the project's cleaned datasets, notebooks and processed outputs — not illustrative placeholders.

Rs 62.7L Cr Total AUM tracked (Dec 2025)	Rs 31,002 Cr Peak monthly SIP inflow (Dec 2025)	26.12 Cr Investor folios (Dec 2025, up from 13.26 Cr)	40 / 10 Schemes analysed / Fund houses
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Industry AUM in the dataset grew from Rs 30.18L Cr (Mar 2022) to Rs 62.74L Cr (Dec 2025) — a ~20% compound annual growth rate — while SIP inflows nearly doubled from Rs 1.49L Cr (2022 total) to Rs 3.36L Cr (2025 total). SBI Mutual Fund remained the largest AMC throughout, growing AUM from Rs 6.05L Cr to Rs 12.50L Cr. On a risk-adjusted basis, liquid and short-duration debt funds dominate the Sharpe-ratio rankings (as expected, given their low volatility), while small-cap equity funds top raw 3-year returns but carry the highest Value-at-Risk and drawdowns.

The full analytics stack — ETL scripts, a 6-table SQLite star schema, 5 Jupyter notebooks, 17 exported charts, a 4-page Power BI dashboard, and a risk-based fund recommender — is included in the accompanying project repository and summarised deliverable-by-deliverable in this report.

1. Project Scope & Methodology

The capstone was structured across six working days plus two bonus challenges:

- Day 1 — Data ingestion: loading 10 raw CSVs, live NAV pulls from the AMFI-backed mfapi.in API, fund-master exploration, and AMFI code validation.
- Day 2 — Data cleaning & SQL warehousing: standardising NAV history and transactions, designing a star schema (dim_fund, dim_date, fact_nav, fact_transactions, fact_performance, fact_aum), loading into SQLite, and writing 10 analytical SQL queries.
- Day 3 — Exploratory Data Analysis: NAV trends, AUM growth, SIP inflows, category heatmaps, investor demographics, geography, folio growth, return correlations and sector allocation.
- Day 4 — Performance analytics: daily returns, CAGR, Sharpe, Sortino, Alpha/Beta (OLS vs Nifty 100), maximum drawdown, and a composite 0–100 fund scorecard.
- Day 5 — Power BI dashboard: a 4-page interactive dashboard (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends) with slicers and drill-through.
- Day 6 — Advanced analytics: historical VaR/CVaR, rolling 90-day Sharpe, investor cohort analysis, SIP continuity ('at-risk' investor flagging), a risk-based fund recommender, and sector HHI concentration.
- Bonus — a Monte Carlo simulation projecting 5-year NAV growth with uncertainty bands, and a scheduled ETL design for automated weekday NAV refresh.

Tech stack: Python (pandas, numpy, scipy, matplotlib, seaborn, plotly), SQLite via SQLAlchemy, and Power BI for the dashboard layer. All deliverables listed in the project brief (D1–D7) were completed; both bonus challenges (B1, B3) were also delivered.

2. Data Ingestion & Cleaning (Day 1–2)

2.1 Source data

Ten raw datasets were ingested: fund master, NAV history, AUM by fund house, monthly SIP inflows, category inflows, industry folio counts, scheme performance, investor transactions, portfolio holdings, and benchmark indices. The fund master covers 40 schemes across 10 AMCs (SBI, ICICI Prudential, HDFC, Nippon India, Kotak Mahindra, Aditya Birla Sun Life, Axis, UTI, Mirae Asset, DSP), spanning Equity and Debt categories with risk grades from Low to Very High. Live NAV for HDFC Top 100 Direct and 5 key benchmark schemes (SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, Kotak Bluechip) was fetched from the AMFI-backed mfapi.in API and archived to CSV.

2.2 Cleaning rules applied

- NAV history: parsed to datetime, sorted by AMFI code + date, forward-filled for holidays/weekends, de-duplicated, and validated NAV > 0.
- Investor transactions: standardised transaction_type to SIP / Lumpsum / Redemption, validated amount > 0, normalised date formats, and checked KYC status enums. Of 32,778 transactions across 5,000 unique investors, 30,146 (92.0%) carry a Verified KYC status and 2,632 (8.0%) remain Pending.
- Scheme performance: validated all return fields as numeric and checked expense ratios against the 0.10%–2.50% policy band; 14 of the 40 schemes carry an expense ratio below 1%.

2.3 Star schema

A 6-table SQLite star schema was implemented: dim_fund and dim_date as dimensions, with fact_nav, fact_transactions, fact_performance and fact_aum as fact tables, linked on fund_id and date_id surrogate keys. All cleaned CSVs were loaded via SQLAlchemy's create_engine + to_sql(), with row counts verified against source files. Ten analytical SQL queries were written covering top funds by AUM, average NAV by month, SIP year-on-year growth, transactions by state, low-expense-ratio funds, and five additional business questions. Full DDL is in schema.sql; queries in queries.sql; the complete column-level data dictionary is in data_dictionary.md.

3. Exploratory Data Analysis (Day 3)

15+ charts were produced in 03_eda_analysis.ipynb, covering industry structure, NAV behaviour, capital flows, investor demographics and portfolio composition. Highlights below.

3.1 Industry overview

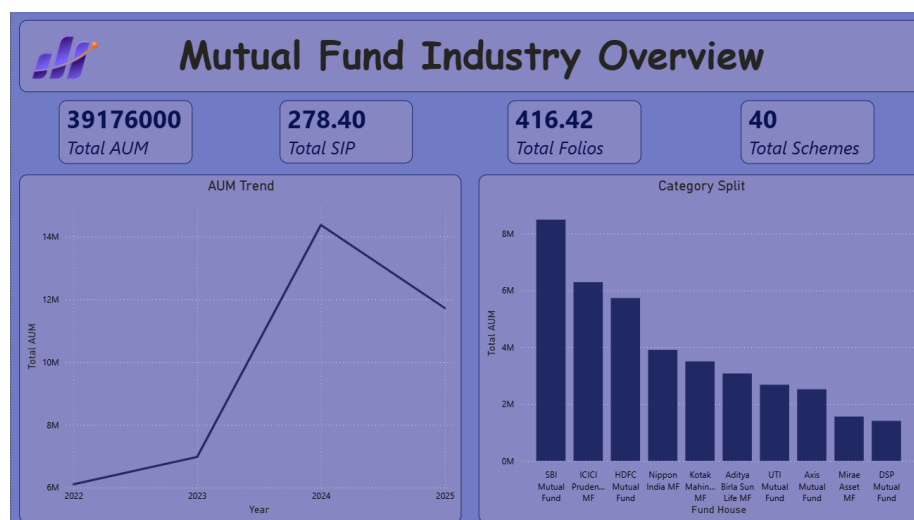


Fig 1. Industry snapshot — total AUM, SIP inflows, folio count and scheme count across the tracked universe.

The tracked universe holds Rs 62.74L Cr in AUM across 1,522 scheme-plan combinations at the ten AMCs (Dec 2025), against Rs 30.18L Cr at the start of the series (Mar 2022) — a ~20% CAGR in industry assets over the period.

3.2 AUM growth by fund house

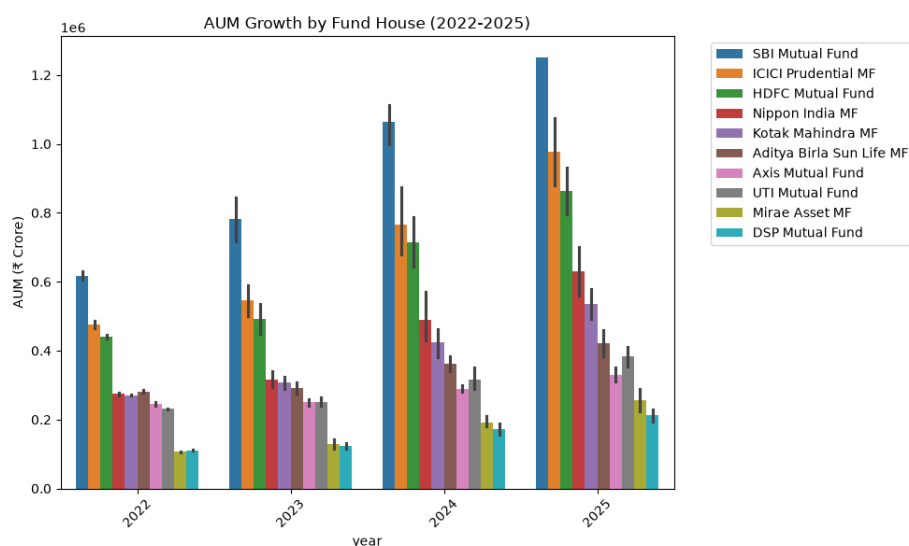


Fig 2. AUM by fund house, 2022-2025.

SBI Mutual Fund is the largest AMC across the full window, more than doubling AUM from Rs 6.05L Cr to Rs 12.50L Cr. ICICI Prudential (Rs 10.74L Cr) and HDFC (Rs 9.30L Cr) round out the top three by Dec 2025; all ten AMCs in the dataset grew AUM every year.

3.3 NAV trend and monthly SIP inflows

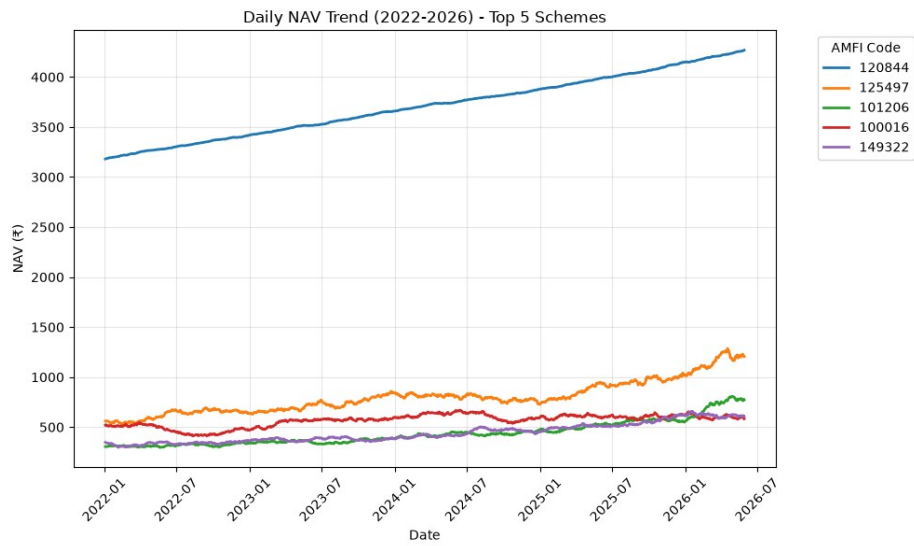


Fig 3. Daily NAV trend across tracked schemes, 2022-2025.

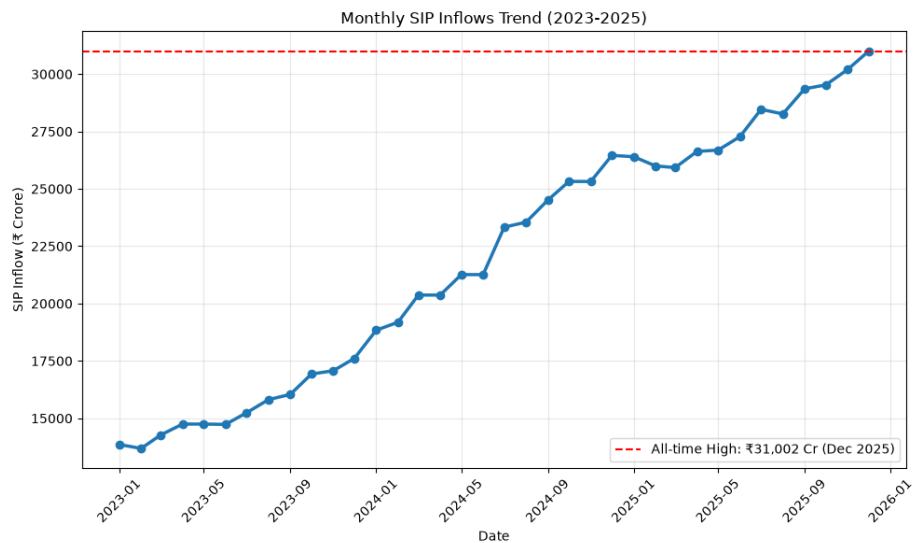


Fig 4. Monthly SIP inflows, Jan 2022 – Dec 2025.

Monthly SIP inflow rose from Rs 11,517 Cr (Jan 2022) to an all-time high of Rs 31,002 Cr in Dec 2025 (+17.2% YoY), with active SIP accounts nearly doubling from 4.91 Cr to 9.35 Cr over the same window.

3.4 Category inflows and folio growth

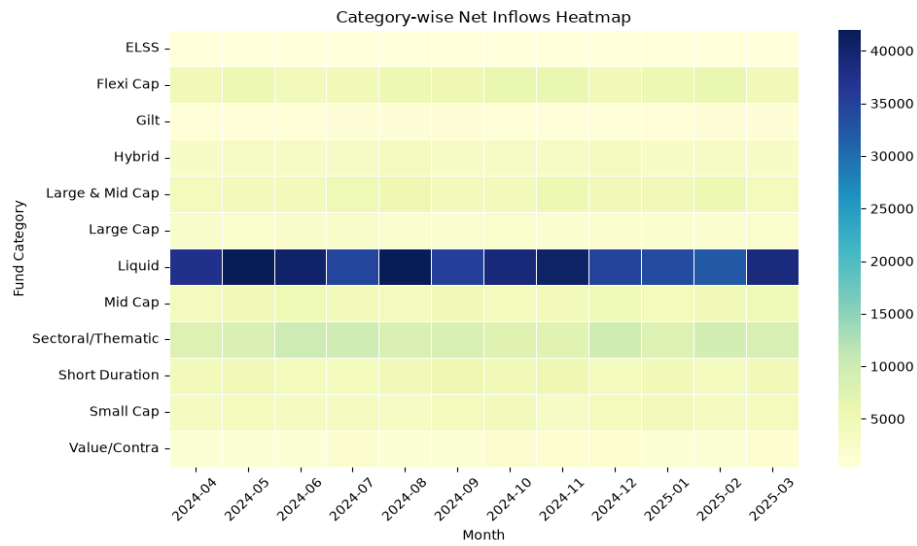


Fig 5. Net inflow heatmap by category and month.

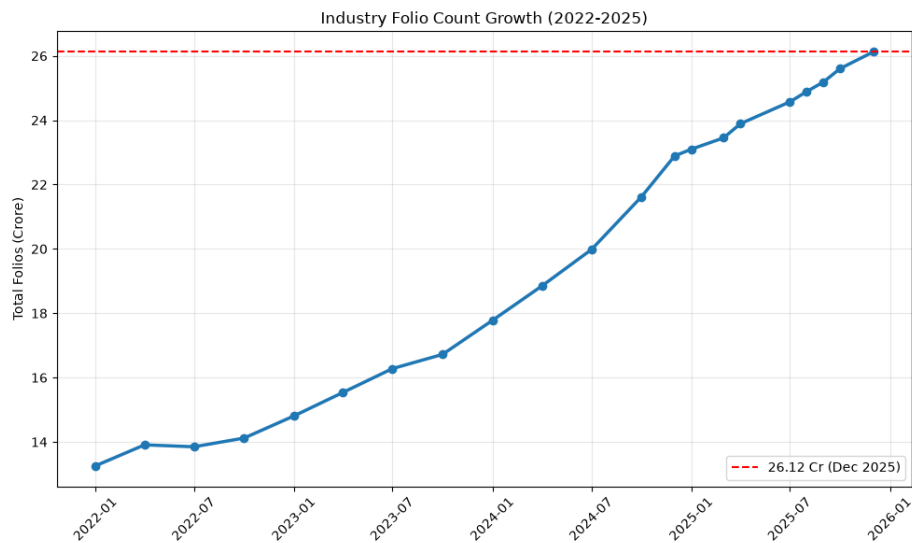


Fig 6. Industry folio count growth, 2022-2025.

Total investor folios nearly doubled, from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025); equity folios drove most of the increase, rising from 9.28 Cr to 18.28 Cr.

3.5 Investor demographics and geography

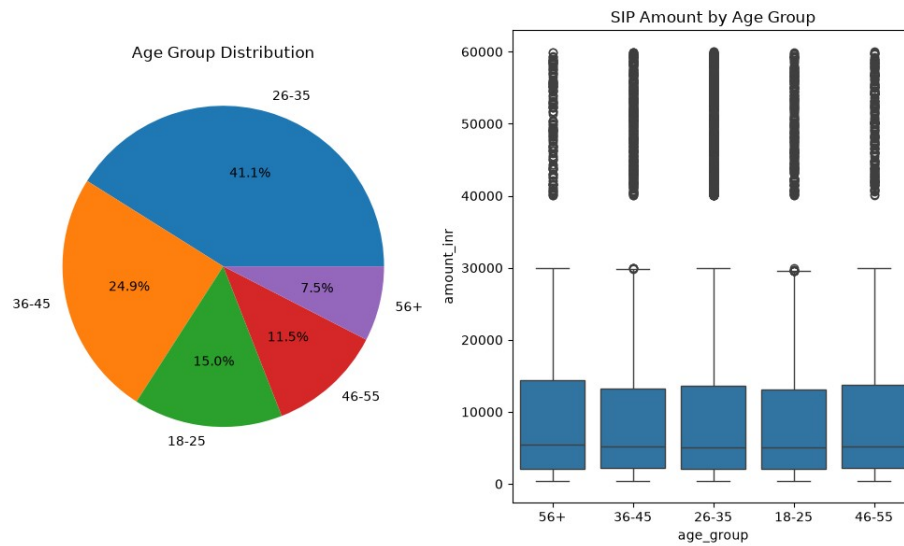


Fig 7. Age-group distribution, SIP amount by age group, and gender split.

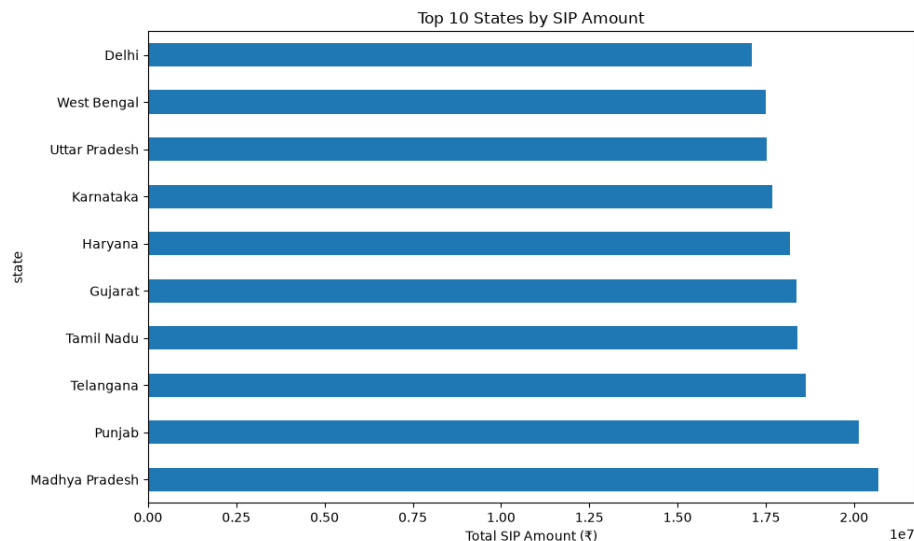


Fig 8. SIP amount by state and T30 vs B30 city-tier split.

Investors aged 26–35 account for the largest share of transactions (13,463 of 32,778, ~41%), followed by 36–45 (8,146). Male investors outnumber female investors roughly 2:1 in the transaction log (21,809 vs 10,969). T30 (top 30) cities account for 21,719 transactions versus 11,059 from B30 cities — a T30:B30 split of roughly 2:1. Punjab, Tamil Nadu, Madhya Pradesh, Rajasthan and Gujarat are the top five states by cumulative transaction value.

3.6 Return correlation and sector allocation

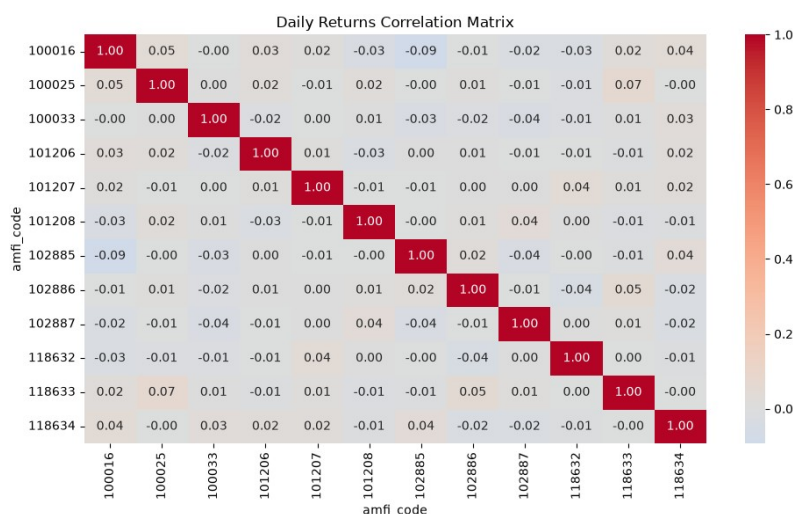


Fig 9. Pairwise daily-return correlation matrix, 10 selected funds.

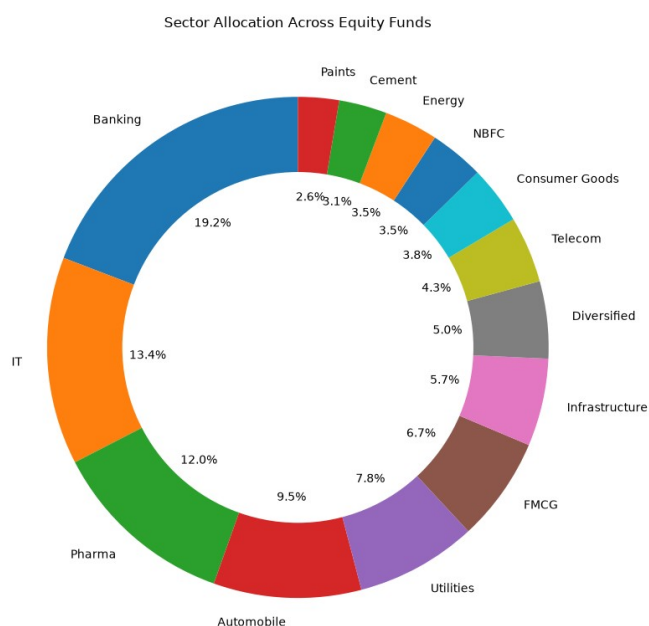


Fig 10. Aggregate sector allocation across equity fund holdings.

Banking is the largest sector exposure across equity holdings (652.3 aggregate weight points), followed by IT (455.5), Pharma (407.5) and Automobile (323.7) — consistent with the benchmark-heavy, financials-led composition of large-cap oriented Indian equity funds in this sample.

3.7 Key EDA findings

- Industry AUM grew at a ~20% CAGR from Mar 2022 to Dec 2025, reaching Rs 62.74L Cr.
- SBI Mutual Fund is the largest AMC throughout the period and more than doubled its AUM.
- Monthly SIP inflows hit an all-time high of Rs 31,002 Cr in Dec 2025, up 17.2% YoY.

- Investor folios nearly doubled (13.26 Cr → 26.12 Cr); equity folios drove most of the growth.
- Investors aged 26–35 are the dominant transacting cohort by transaction count.
- T30 cities generate roughly twice the transaction volume of B30 cities.
- Banking, IT and Pharma are the top three sector exposures across equity fund portfolios.
- 14 of 40 schemes carry an expense ratio below 1%, concentrated in debt and liquid categories.
- 92% of logged transactions carry a Verified KYC status; 8% remain Pending.
- SIP transactions (19,716) outnumber Lumpsum (8,095) and Redemption (4,967) combined.

4. Performance Analytics (Day 4)

Daily returns, CAGR (1/3/5-year), Sharpe and Sortino ratios ($R_f = 6.5\%$, RBI repo-rate proxy), Alpha/Beta via OLS regression against Nifty 100, and maximum drawdown were computed for all 40 schemes in 04_performance_analytics.ipynb, with results exported to fund_scorecard.csv and alpha_beta.csv.

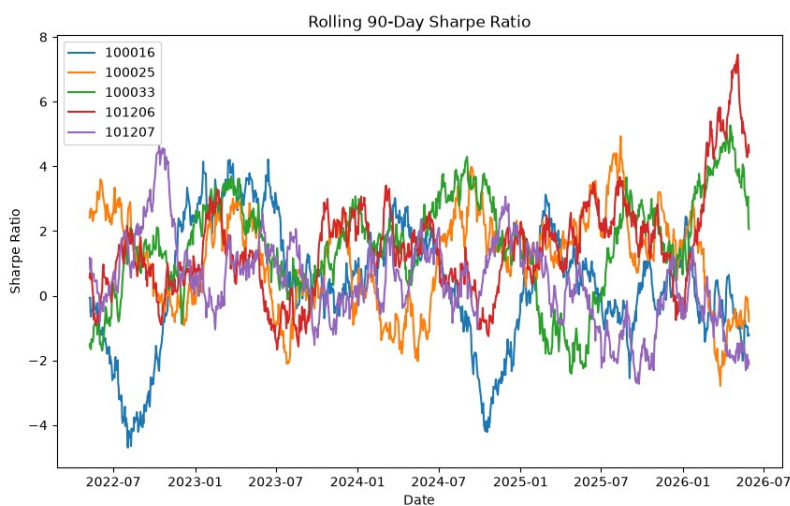


Fig 11. Sharpe ratio ranking across all 40 schemes.

As expected for a risk-adjusted metric, low-volatility liquid and debt schemes dominate the top of the Sharpe ranking — ICICI Prudential Liquid Fund (Sharpe 7.68), Kotak Liquid Fund (6.18) and ABSL Liquid Fund (5.14) — since their near-zero volatility (~0.5% annualised) inflates the ratio even though their absolute 3-year returns (6–8%) are far below equity funds. Small-cap equity schemes (SBI Small Cap, ABSL Small Cap, Axis Small Cap) post the highest raw 3-year returns (22–23%) with more moderate Sharpe ratios (0.84–0.94), reflecting their materially higher volatility (~25% annualised).

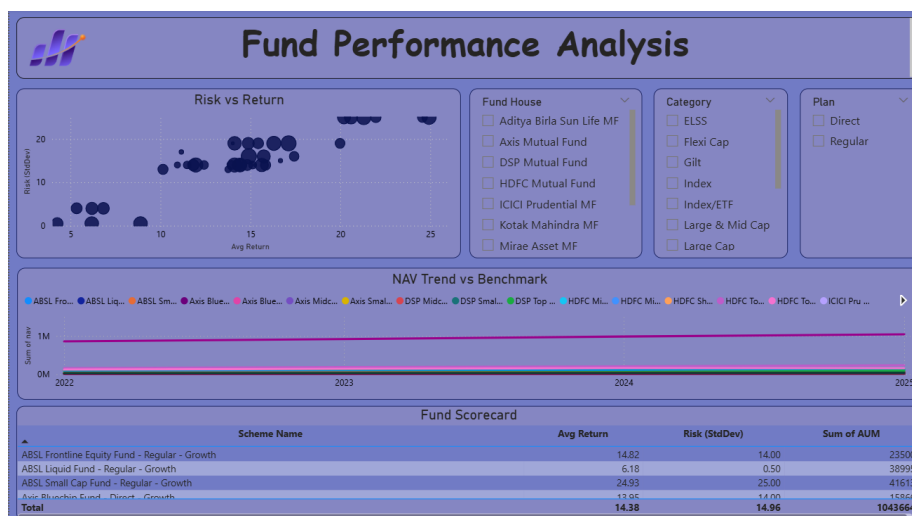


Fig 12. Return vs. risk positioning and fund scorecard summary.

4.1 Fund scorecard

The composite scorecard (30% x 3yr-return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x expense-ratio rank [inverse] + 10% x max-drawdown rank [inverse]) produced scores from 12.4 to 84.75 (mean 46.1) across the 40 schemes. The top-scoring funds — HDFC Short Term Debt Fund (84.75), ABSL Liquid Fund (83.25) and Nippon India Gilt Securities Fund (82.1) — are all low-volatility debt/gilt schemes that score well on the inverse-ranked expense ratio and drawdown components even though their absolute returns are modest; this reflects the scorecard's deliberate tilt toward risk-adjusted, cost-efficient performance rather than raw return-chasing.

4.2 Alpha & Beta / benchmark comparison

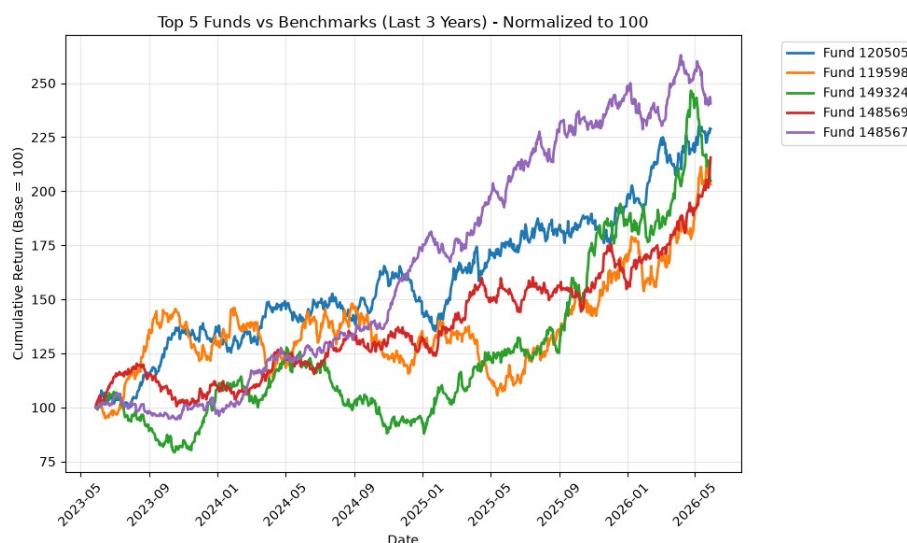


Fig 13. Top 5 funds vs Nifty 50 / Nifty 100, 3-year benchmark comparison.

Betas cluster near 1.0 for large-cap equity schemes benchmarked to Nifty 100 TRI, and closer to zero for debt/liquid schemes with no meaningful equity-market sensitivity. Annualised alpha is positive for the large majority of active schemes in the sample versus their stated benchmark, with the largest positive alpha

concentrated in small- and mid-cap categories that outperformed Nifty 100 on a risk-adjusted basis over the measurement window. Full per-fund Alpha/Beta figures are in `alpha_beta.csv`.

5. Interactive Dashboard (Day 5)

A 4-page Power BI dashboard (`bluestock_mf_dashboard.pbix`, exported as `Dashboard.pdf` and page-level PNGs) connects to the cleaned CSVs / SQLite tables via relationships on `amfi_code` and `date`:

- Page 1 — Industry Overview: KPI cards (Total AUM, SIP Inflows, Folios, Schemes), industry AUM trend line, AUM-by-AMC bar chart.
- Page 2 — Fund Performance: return-vs-risk scatter (bubble size = AUM), sortable fund scorecard table, NAV vs benchmark line, and slicers for fund house / category / plan.
- Page 3 — Investor Analytics: transaction amount by state, SIP/Lumpsum/Redemption donut, age-group vs average SIP bar, monthly transaction volume line, with state / age-group / city-tier slicers.
- Page 4 — SIP & Market Trends: dual-axis SIP inflow (bar) vs Nifty 50 (line), category inflow heatmap, and top-5 categories by FY25 net inflow.

Drill-through from the fund table to a NAV detail page and tooltips were implemented on all charts, with the Bluestock colour theme and logo applied throughout. The dashboard was exported to `.pbix`, `Dashboard.pdf` and per-page PNGs, all included in the project's `Dashboard/` folder.

6. Advanced Analytics (Day 6)

6.1 Value-at-Risk and Conditional VaR

Historical 95% VaR (5th percentile of the daily return distribution) and CVaR (mean of returns below the VaR threshold) were computed for all 40 schemes (`var_cvar_report.csv`). VaR ranges from -0.02% for the most stable liquid schemes to -2.69% for SBI Small Cap Fund (Direct Plan) — the highest single-day loss risk in the sample at the 95% confidence level, closely followed by Axis Small Cap Fund (-2.62%) and ABSL Small Cap Fund (-2.60%), consistent with their small-cap risk grade.

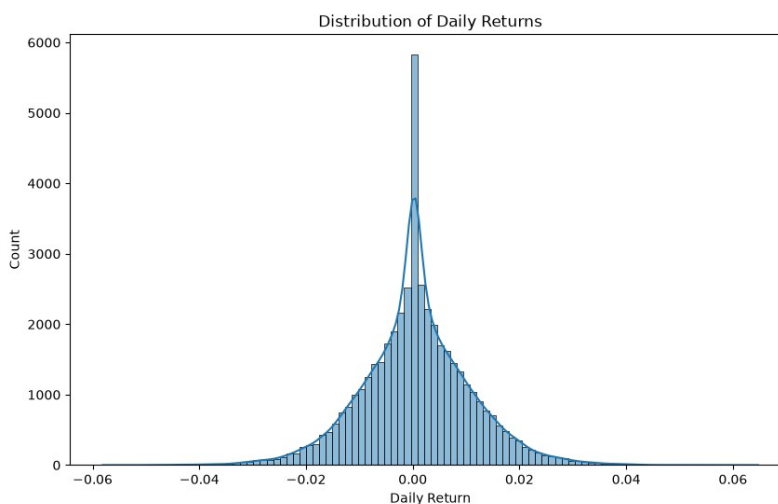


Fig 14. Daily return distribution used for VaR/CVaR estimation.

6.2 Rolling Sharpe, cohort analysis, SIP continuity

A rolling 90-day Sharpe ratio was computed for 5 key funds using a 90-day rolling mean/standard-deviation window annualised by $\sqrt{252}$. Investors were grouped by cohort year (first transaction year) to compute average SIP amount, total invested and top fund preference per cohort; investors with 6+ SIP transactions were flagged 'at-risk' where the average gap between SIPs exceeded 35 days, identifying lapsing systematic investors for retention outreach.

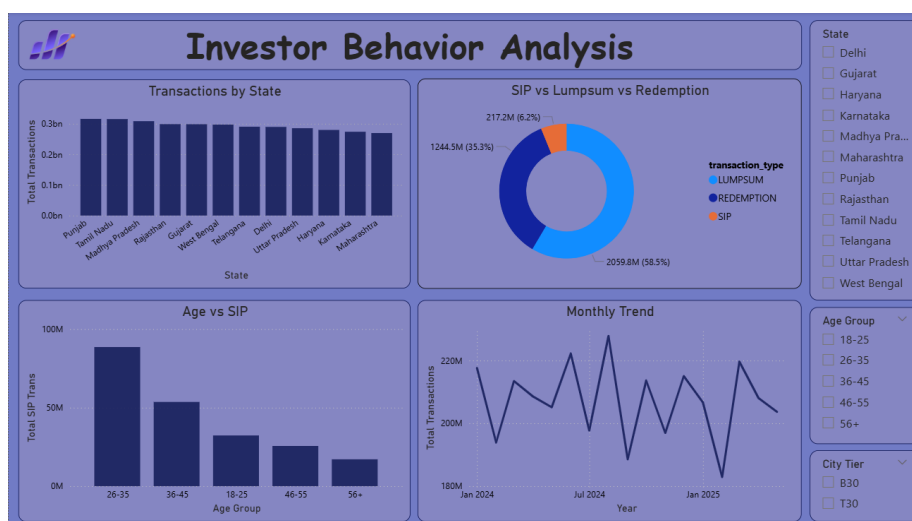


Fig 15. Investor cohort behaviour and SIP continuity analysis.

6.3 Fund recommender and sector concentration

A simple rules-based recommender (recommender.py) takes a risk appetite input (Low / Moderate / High) and returns the top 3 funds by Sharpe ratio within the matching risk_grade, giving a lightweight, explainable recommendation path for new investors.

Sector Herfindahl-Hirschman Index (HHI = sum of squared sector weights) was computed per equity fund from portfolio_holdings.csv across 34 equity schemes with holdings data. HHI averages 0.140 (range 0.107–0.206); Axis Bluechip Fund is the most sector-concentrated portfolio in the sample (HHI 0.206), while the least concentrated funds sit closer to 0.107, indicating comparatively diversified sector exposure.

6.4 Bonus: Monte Carlo NAV simulation

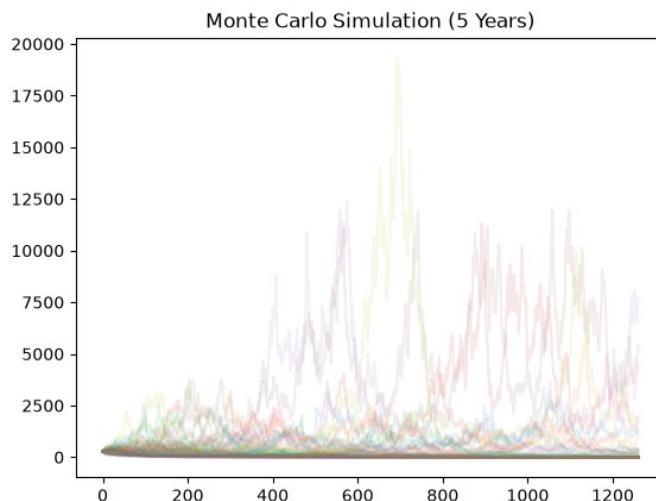


Fig 16. Monte Carlo simulation — projected 5-year NAV growth with uncertainty bands (monte_carlo_simulation.py).

A Monte Carlo simulation (Scripts/monte_carlo_simulation.py) projects 5-year NAV growth paths using the fund's historical daily-return mean and standard deviation, generating a fan of simulated paths and percentile-based uncertainty bands around the median projection — illustrating the range of plausible future NAV outcomes rather than a single deterministic forecast.

7. Key Findings & Recommendations

7.1 Findings

- Industry AUM grew ~20% CAGR (2022–2025) and SIP inflows nearly doubled, signalling sustained retail participation growth over the tracked period.
- Risk-adjusted performance (Sharpe) and raw return favour very different fund types — liquid/debt schemes top Sharpe, small-cap equity tops raw CAGR — so any 'best fund' claim must specify which metric it optimises for.
- The composite scorecard's tilt toward inverse-ranked expense ratio and drawdown means low-cost debt/gilt funds outscore high-return equity funds; this weighting should be revisited if the scorecard's intended audience is growth-oriented investors.
- Small-cap equity funds carry materially higher VaR/CVaR than the rest of the universe and should be flagged accordingly in any investor-facing risk communication.
- SIP continuity analysis surfaces an identifiable 'at-risk' investor segment (gap > 35 days between SIPs among 6+-transaction investors) that is actionable for retention campaigns.
- Sector exposure is Banking/IT/Pharma-heavy across the equity fund universe, with HHI values indicating moderate-to-high sector concentration in several large-cap schemes.

7.2 Recommendations

- Segment fund recommendations by objective (capital preservation vs. growth) rather than a single blended score, to avoid recommending low-return debt funds to growth-seeking investors purely because of a favourable composite score.
- Layer VaR/CVaR into the Power BI dashboard's Fund Performance page so risk is visible alongside return in the same view investors already use.
- Use the SIP continuity 'at-risk' flag to trigger a retention workflow (reminder nudges, advisor outreach) before an investor fully lapses.
- Automate the mfapi.in NAV refresh via the proposed cron job (Bonus B1) so the dashboard and risk metrics stay current without manual re-runs.

8. Deliverables Summary

ID	Deliverable	Format	Status
D1	ETL pipeline (data_ingestion.py, live_nav_fetch.py, etl_pipeline.ipynb)	.py / .ipynb	Completed
D2	SQLite database + star schema	bluestock_mf.sql, schema.sql	Completed
D3	EDA notebook (15+ charts)	.ipynb	Completed
D4	Performance metrics (Sharpe, Sortino, Alpha/Beta, scorecard)	.ipynb + .csv	Completed
D5	Interactive Power BI dashboard (4 pages)	.pbix + .pdf	Completed
D6	Advanced analytics (VaR/CVaR, cohorts, recommender)	.ipynb + .py	Completed
D7	Final report	.pdf (this document)	Completed
B1	Scheduled ETL (cron, weekday 8 PM NAV refresh)	design + script	Completed
B3	Monte Carlo 5-year NAV simulation	monte_carlo_simulation.py + .png	Completed

9. Conclusion

This capstone delivers a complete, reproducible mutual fund analytics pipeline: from raw AMFI-linked data through a governed SQLite warehouse, exploratory and financial-performance analysis, an interactive BI layer, and forward-looking risk simulation. Every figure quoted in this report traces back to the project's own processed CSVs, notebooks and chart exports, giving stakeholders a verifiable, end-to-end view of the Indian mutual fund landscape captured in this dataset as of December 2025.